

Chayut Wongkamthong

(+66) 819632331 | chayutwo@gmail.com | chayutwo.github.io

 [chayut-wongkamthong](#) |  [ChayutWo](#)

Bangkok, Thailand

SUMMARY

Data scientist and applied researcher with years of experience across banking, energy, and academia. Leads a nine-person data science team delivering production AI in credit risk, personalization, optimization, and LLM applications, supported by peer-reviewed research spanning statistics, signal processing, and applied machine learning.

EXPERIENCE

- **KASIKORN Business-Technology Group | KBTG**  Bangkok, Thailand
Senior Data Scientist (Data Science Squad Leader) 2026 - Present
Advanced Data Scientist 2022 - 2025
 - **Technical Leadership:** Lead nine data scientists delivering production AI across credit risk, debt collection, personalization, and wealth management at one of Thailand's largest banks.
 - **Applied Research:** Translate data science and optimization research into banking products spanning dynamic pricing, asset valuation, infrastructure planning, personalization, and LLMs.
- **Duke University**  NC, USA
Research Assistant – Concurrent Appointments in SSRI and ECE Jan. 2020 - Dec. 2021
 - **Social Science Research Institute (SSRI)** : Researched Bayesian methods for missing data imputation and Dirichlet process mixture models under Assistant Professor Olanrewaju M. Akande.
 - **Electrical and Computer Engineering (ECE)** : Researched deep learning models and statistical simulation methods for radar object detection and cluster analysis under Prof. Vahid Tarokh and Prof. Ali Pezeshki.
- **PTT Exploration and Production Public Company Limited (PTTEP)**  Bangkok, Thailand
Petroleum Engineer Jul. 2015 - Aug. 2019
 - **JV Technical Lead with Chevron:** Led technical coordination for the B8/32 offshore oil field in the Gulf of Thailand.
 - **Reservoir Engineer:** Optimized production planning for Arthit, a 220-MMcf-per-day gas field.

PROJECTS

- **AI-Powered Market Research Platform for Credit Card Promotions** Jan. 2026 - Present
Tools: LLM Agents, Adaptive Web Crawling, Information Extraction, RAG - Python
 - Spearheaded an AI-powered pipeline to automatically crawl, extract, and analyze banking products and credit card promotions across major Thai banks.
 - Automated competitive analysis of credit card promotions through a centralized dashboard, saving the business team 15 person-days per month.
 - Published an IEEE paper on LLM-guided adaptive web crawling in the JCSSE 2026 proceedings.
- **Location- and Spending-Triggered Credit Card Promotion Recommendation** Mar. 2025 - Jun. 2026
Tools: Recommendation Systems, Geospatial Analytics, Behavioral Modeling - Python
 - Engineered a real-time recommendation system using spending and location triggers to deliver personalized credit card promotions through K PLUS.
 - Diagnosed and resolved recommendation drop-offs across POI boundaries, filters, campaign curation, and model mismatches, adding 30,000 daily deliveries and cutting drop-offs by over 50%.
 - Validated impact through A/B testing: THB 2,500 higher monthly spending per targeted customer, projected to THB 1.8 billion in monthly incremental spending across the eligible customer base.
- **Transaction Load Prediction for Infrastructure Capacity Planning** Jan. 2025 - May 2026
Tools: Time-Series Forecasting, ARIMA, N-BEATS, Chronos - Python, R
 - Replaced a regression model with time-series forecasting models, reducing prediction error by 18% and providing evidence that helped avoid premature infrastructure investment.

- Enhanced server-load monitoring routines and trained the IT team to operate and maintain them.
- **Best Time to Call Model and Optimization for Debt Collection Process** Nov. 2023 - Dec. 2024
Tools: Multi-Armed Bandits, Numerical Optimization - Python
 - Led a team of six to formulate the collection optimization task as a multi-armed bandit problem, implementing Top-Two Thompson Sampling to determine the best time to call each debtor.
 - Optimized KBank's debt collection pipeline, achieving a 16% relative increase in contact rate, THB 350,000 in cost savings, and over THB 12 million in additional monthly collections.
- **NPA Land Valuation Using Satellite Imagery** Jun. 2023 - Nov. 2024
Tools: Computer Vision, Remote Sensing, Machine Learning - Python
 - Designed a satellite-imagery-based model to estimate land values for non-performing assets.
 - Expanded valuation coverage by 10% and published the methodology in the IEEE TENCON 2023 proceedings.
- **NPA Liquidity Modeling for Optimal Pricing** Apr. 2023 - Jul. 2023
Tools: Classification, Asset Segmentation, Dynamic Pricing - Python
 - Developed a classification model to segment non-performing assets into high-, medium-, and low-liquidity groups with 96% accuracy.
 - Deployed the model to support liquidity-based pricing decisions during a year in which NPA sales reached a record THB 1.3 billion; the project received KBank's 2023 Breakthrough Award.
- **Data-Driven Improved Radar Object Detection** Jan. 2020 - Dec. 2021
Tools: Deep Learning, Signal Processing - MATLAB, Python
 - Developed and benchmarked statistical and deep-learning methods for radar clutter cancellation and object detection in collaboration with the Air Force Research Laboratory.
 - Co-authored papers published in the proceedings of the 2021 and 2022 IEEE Radar Conferences.
- **Statistical and Machine Learning Methods for Imputing Ordinal Data** Jan. 2020 - Nov. 2020
Tools: Missing Data Imputation, Generative Adversarial Networks (GANs) - R, Python 
 - Compared statistical and machine-learning methods for ordinal-data imputation, including Dirichlet process models and MICE, evaluating inferential validity and distributional fidelity.
 - Published the findings in the Journal of Survey Statistics and Methodology (JSSAM).
- **Production Optimization Software: The Field State Model** Jan. 2018 - Oct. 2018
Tools: Convex Optimization - MATLAB 
 - Developed software to formulate the petroleum production system as a convex optimization problem.
 - Partnered with asset managers to deploy the software, generating over \$640,000 in incremental production value in 2018.
 - Published and presented the work at the 2018 SPE Asia Pacific Oil and Gas Conference and Exhibition in Brisbane, Australia.

SKILLS

- **Programming Languages:** Python, SQL, R, MATLAB
- **Data Science and AI:** Machine Learning, Bayesian Modeling, Reinforcement Learning, Numerical Optimization, Time-Series Forecasting, and LLM Engineering
- **AI Engineering and Infrastructure:** LangGraph, LangChain, PyTorch, Google Cloud Platform, Vertex AI, MongoDB, OpenSearch, and Git

EDUCATION

- **Duke University** Aug. 2019 - May 2021
M.S. in Data Science NC, USA
 - GPA: 3.97/4.00
- **Chulalongkorn University** May 2011 - Jun. 2015
Bachelor of Engineering Bangkok, Thailand
 - GPA: 4.00/4.00, First Class Honors

[C.5] C. Anaroch, P. Choobanna, S. Boonaneksap, B. Changaival, and C. Wongkamthong, **LLM-Guided Adaptive Web Crawling: A Case Study in Banking Products**, *Proc. 2026 23rd International Joint Conference on Computer Science and Software Engineering (JCSSE)*, Bangkok, Thailand, pp. 805–810, 2026, doi: 10.1109/JCSSE68839.2026.11596755.

[P.1] C. Wongkamthong, **Hierarchical Dirichlet Process Mixture of Products of Multinomial Distributions: Applications to Survey Data with Potentially Missing Values**, *arXiv preprint arXiv:2412.17335*, 2024.

[C.4] S. Puengdang, W. Ausawalaithong, P. Nopratana Wong, N. Keeratipranon, and C. Wongkamthong, **Thailand Asset Value Estimation Using Aerial or Satellite Imagery**, *Proc. 2023 IEEE Region 10 Conference (TENCON)*, Chiang Mai, Thailand, pp. 399–404, 2023, doi: 10.1109/TENCON58879.2023.10322494.

[J.1] C. Wongkamthong and O. Akande, **A Comparative Study of Imputation Methods for Multivariate Ordinal Data**, *Journal of Survey Statistics and Methodology*, vol. 11, no. 1, pp. 189–212, 2023, doi: 10.1093/jssam/smab028.

[C.3] S. Venkatasubramanian, C. Wongkamthong, M. Soltani, et al., **Toward Data-Driven STAP Radar**, *Proc. 2022 IEEE Radar Conference (RadarConf22)*, New York City, NY, USA, pp. 1–5, 2022, doi: 10.1109/RadarConf2248738.2022.9764354.

[C.2] Y. Feng, C. Wongkamthong, M. Soltani, et al., **Knowledge-Aided Data-Driven Radar Clutter Representation**, *Proc. 2021 IEEE Radar Conference (RadarConf21)*, Atlanta, GA, USA, pp. 1–4, 2021, doi: 10.1109/RadarConf2147009.2021.9455318.

[C.1] C. Wongkamthong, K. Wongpattananukul, C. Suranetinai, et al., **In-House Software Development for Gas Production Optimization: A South East Asia Perspective**, *Proc. 2018 SPE Asia Pacific Oil and Gas Conference and Exhibition*, Brisbane, Australia, 2018, doi: 10.2118/192080-MS.

HONORS AND AWARDS

• Duke University Scholarship <i>The Social Science Research Institute (SSRI), Duke University</i>	2019
• First Class Honors and Medal of Excellence <i>Chulalongkorn University</i>	2015
• King Bhumibol Scholarship <i>HM King Rama IX</i>	2015
• Gold Medal for Academic Achievement <i>King Rama X through the Engineering Institute of Thailand</i>	2014

CERTIFICATIONS

• Anthropic: Claude Certified Architect – Foundations	Jul. 2026
• Google Skills: Completed AI Research Foundations and Generative AI for Developers learning paths	2026
• Google Cloud on Coursera: Generative AI Leader	Feb. 2026
• Google on Coursera: Google People Management Essentials	Jan. 2026
• University of Alberta & Amii on Coursera: Reinforcement Learning Specialization Certificate	Mar. 2024
• University of Virginia & BCG on Coursera: Pricing Strategy Optimization Certificate	Feb. 2024
• DeepLearning.AI: Deep Learning Specialization Certificate	Sep. 2019
• IBM on Coursera: IBM Data Science Professional Certificate	Apr. 2019